

Work and kinetic energy

Solving a general problem:

Given $\vec{F}(\vec{r}, \vec{v}, t)$, solve $\vec{r}(t)$

Euler integration

step	time	position	velocity	acceleration
0	t_0	$\vec{r}_0$	$\vec{v}_0$	$\vec{a}_0 = \vec{F}(\vec{r}_0, \vec{v}_0, t_0)/m$
1	$t_1 = t_0 + \Delta t$	$\vec{r}_1 = \vec{r}_0 + \vec{v}_0 \Delta t$	$\vec{v}_1 = \vec{v}_0 + \vec{a}_0 \Delta t$	$\vec{a}_1 = \vec{F}(\vec{r}_1, \vec{v}_1, t_1)/m$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$t_n = t_{n-1} + \Delta t$	$\vec{r}_n = \vec{r}_{n-1} + \vec{v}_{n-1} \Delta t$	$\vec{v}_n = \vec{v}_{n-1} + \vec{a}_{n-1} \Delta t$	$\vec{a}_n = \vec{F}(\vec{r}_n, \vec{v}_n, t_n)/m$

• Error of $\vec{r}$: $O(\Delta t^2)$

Convergence check: $\Delta t' = \frac{1}{2} \Delta t, \frac{1}{4} \Delta t, \dots$

Verlet integration

$$\vec{r}_{n+1} = 2\vec{r}_n - \vec{r}_{n-1} + \vec{a}_n \Delta t^2$$

Proof:
$$\begin{cases} \vec{r}_{n+1} - \vec{r}_n = \cancel{\vec{v}_n \Delta t} + \frac{1}{2} \vec{a}_n \Delta t^2 + \cancel{\frac{1}{6} \vec{b}_n \Delta t^3} + O(\Delta t^4) \\ \vec{r}_n - \vec{r}_{n-1} = \cancel{-\vec{v}_{n-1} \Delta t} + \frac{1}{2} \vec{a}_n \Delta t^2 - \cancel{\frac{1}{6} \vec{b}_{n-1} \Delta t^3} + O(\Delta t^4) \end{cases} \quad \vec{b} \equiv \frac{d\vec{a}}{dt}$$

• Error of $\vec{r}$: $O(\Delta t^4)$!!!

• $\vec{r}_1$ required to start the iteration:

$$\vec{r}_1 \approx \vec{r}_0 + \vec{v}_0 \Delta t + \frac{1}{2} \vec{a}_0 \Delta t^2 \quad O(\Delta t^3)$$

• Note: $\vec{v}_n$ not required for $n > 1$.

Velocity Verlet integration

assume $\vec{F}(\vec{r}, t)$, no dependence on $\vec{v}$

$$\textcircled{1} \quad \vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2} \vec{a}_n \Delta t^2$$

$$\textcircled{2} \quad \vec{a}_{n+1} = \vec{F}(\vec{r}_{n+1}, t_{n+1})/m$$

①

$$\textcircled{3} \quad \vec{v}_{n+1} = \vec{v}_n + \frac{1}{2} (\vec{a}_n + \vec{a}_{n+1}) \Delta t$$

- Error $\mathcal{O}(\Delta t^4)$
- Only depends on the previous step
- More compact form of velocity Verlet integration:

$$\text{Define } U_F(\Delta t) \begin{pmatrix} \vec{r} \\ \vec{v} \end{pmatrix} = \begin{pmatrix} \vec{r} \\ \vec{v} + \vec{a} \Delta t \end{pmatrix}$$

$$U_V(\Delta t) \begin{pmatrix} \vec{r} \\ \vec{v} \end{pmatrix} = \begin{pmatrix} \vec{r} + \vec{v} \Delta t \\ \vec{v} \end{pmatrix}$$

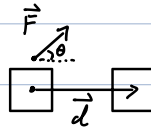
$$\begin{aligned} \Rightarrow U_F\left(\frac{1}{2}\Delta t\right) U_V(\Delta t) U_F\left(\frac{1}{2}\Delta t\right) \begin{pmatrix} r_n \\ v_n \end{pmatrix} \\ = U_F\left(\frac{1}{2}\Delta t\right) U_V(\Delta t) \begin{pmatrix} r_n \\ v_n + \frac{1}{2}a_n \Delta t \end{pmatrix} \rightarrow v'_{n+1} \\ = U_F\left(\frac{1}{2}\Delta t\right) \begin{pmatrix} r_n + (v_n + \frac{1}{2}a_n \Delta t) \Delta t \\ v_n + \frac{1}{2}a_n \Delta t \end{pmatrix} \rightarrow r_{n+1} \Rightarrow a_{n+1} \text{ can be calculated} \\ \rightarrow v'_{n+1} \\ = \begin{pmatrix} r_n + (v_n + \frac{1}{2}a_n \Delta t) \Delta t \\ v_n + \frac{1}{2}a_n \Delta t + \frac{1}{2}a_{n+1} \Delta t \end{pmatrix} \rightarrow r_{n+1} \\ \rightarrow v_{n+1} \end{aligned}$$

- Work

$$W = F d \cos\theta = \vec{F} \cdot \vec{d}$$

$\vec{F}$: constant force

$\vec{d}$: displacement



Notes:

$$\textcircled{1} \quad \vec{d} = 0 \Rightarrow W = 0 \text{ even for } \vec{F} \neq 0$$

$$\textcircled{2} \quad \vec{F} \perp \vec{d}$$

$$\begin{array}{c} \uparrow \vec{F} \\ \square \\ \rightarrow \vec{d} \end{array} \Rightarrow W = 0$$

Unit: joule (J) = N · m

$\textcircled{2}$

Another definition of "work":

energy transferred between an object and force.

positive work $\leftrightarrow$ transferred to object

negative work $\leftrightarrow$ transferred from object.

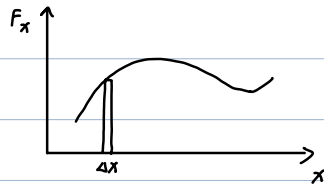
$$W = \vec{F} \cdot \vec{d}$$

reminder on scalar product:

$$\left. \begin{aligned} \vec{A} &= \sum_i A_i \vec{e}_i \\ \vec{B} &= \sum_j B_j \vec{e}_j \end{aligned} \right\} \Rightarrow \vec{A} \cdot \vec{B} = \sum_i \sum_j A_i B_j \vec{e}_i \cdot \vec{e}_j$$

for orthonormal basis $\vec{e}_i \cdot \vec{e}_j = \delta_{ij} \Rightarrow \vec{A} \cdot \vec{B} = \sum_i A_i B_i$
"正交" "正交"

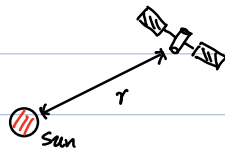
• Varying Force,



For $\Delta x \rightarrow 0$: $\Delta W = F_x \Delta x$

$$\Rightarrow W = \int_{x_i}^{x_f} F_x dx \quad \rightarrow \text{area covered by } F_x$$

Example - I. Gravitational force



$$F = -1.3 \times 10^{22} / r^2$$

$$r_i: 1.5 \times 10^{11} \text{ m}$$

$$r_f: 2.3 \times 10^{11} \text{ m}$$

$$\begin{aligned} W &= \int_{r_i}^{r_f} dr \left(\frac{-1.3 \times 10^{22}}{r^2} \right) \\ &= 1.3 \times 10^{22} \left. \frac{1}{r} \right|_{r_i}^{r_f} \\ &= 1.3 \times 10^{22} \left(\frac{1}{2.3 \times 10^{11}} - \frac{1}{1.5 \times 10^{11}} \right) \\ &\approx -3 \times 10^{10} \text{ J} \end{aligned}$$

Example-II Hooke's Law (force of springs)

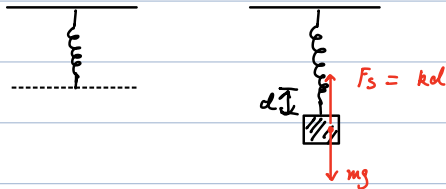
$$F_s = -kx$$

k : force constant

(弹性系数
刚性系数
劲度系数)

x : displacement from relaxed position

Measure k :

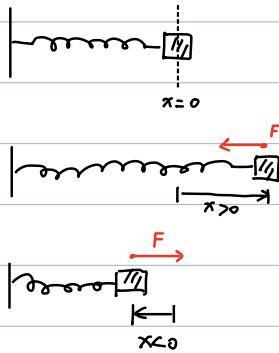


(equilibrium condition)

$$kd = mg \Rightarrow k = mg/d$$

Notes:

• "-" sign: force always opposite to displacement.



Work by spring:

$$\begin{aligned} W &= \int_{x_i}^{x_f} -kx \, dx \\ &= -\frac{1}{2}k(x_f^2 - x_i^2) \end{aligned}$$

e.g. ① $x_i = 0$, $x_f = x_{\max}$

$$\Rightarrow W = -\frac{1}{2}kx_{\max}^2$$

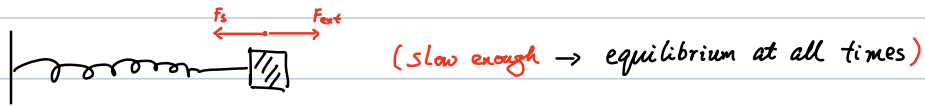
e.g. ② $x_i = 0$, $x_f = -x_{\max}$

$$\Rightarrow W = -\frac{1}{2}kx_{\max}^2 \quad (\text{SAME sign})$$

e.g. ③ $x_i = -x_{\max}$, $x_f = x_{\max}$

$\Rightarrow W = 0$ (回到解放前)

Work by *slow* external force:



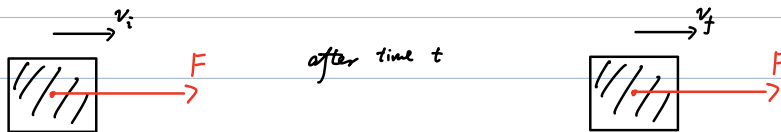
$$F_{\text{ext}} = -F_s$$

$$\Rightarrow W_{\text{ext}} = -W_s$$

• Kinetic Energy

Q: Why define $W = \vec{F} \cdot \vec{d}$?

Consider constant acceleration



$$v_f = v_i + at$$

$$a = F/m$$

$$d = v_i t + \frac{1}{2} at^2$$

$$= \frac{1}{2} (v_i + v_f) t$$

$$\text{Cancel } t \Rightarrow v_f = v_i + a \cdot \frac{2d}{v_i + v_f}$$

$$\Rightarrow \frac{1}{2} (v_f^2 - v_i^2) = ad$$

$$\downarrow$$

$$\frac{F}{m}$$

$$\Rightarrow \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = Fd$$

change of kinetic energy $\downarrow$ Work

$K \equiv \frac{1}{2} m v^2$

Work - kinetic energy theorem

$$K_f - K_i = W \quad \text{for } \forall F \quad (\text{even non-constant})$$

Proof:

$$\begin{aligned} W &= \int_{x_i}^{x_f} F \, dx \\ &= \int_{x_i}^{x_f} ma \, dx \\ &= \int_{x_i}^{x_f} m \frac{dv}{dt} \, dx \\ &= \int_{x_i}^{x_f} m v \, dv \\ &= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \end{aligned}$$

• Power : time rate of work done

Average power:

$$P_{\text{avg}} = \frac{W}{\Delta t}$$

Instantaneous power:

$$P = \frac{dW}{dt}$$

for constant force:

$$P = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

Unit:

$$1 \, \text{W} = \frac{1 \, \text{J}}{1 \, \text{s}} = 1 \, \text{kg} \cdot \text{m}^2 / \text{s}^3 \quad (\text{watt})$$

(James Watt, 1736 - 1819)

$$\text{kilowatt-hour: } \text{kWh} = 1 \, \text{kW} \cdot 1 \, \text{h}$$

$$= 10^3 \, \text{W} \cdot 3600 \, \text{s}$$

$$= 3.6 \times 10^6 \, \text{J}$$